WEEK 10 | OBJECT DETECTION II

YCB-V full-frame detection • Confidence, NMS and Precision–Recall

Intern	Main Quest	Group	Side Quest
Mehrana Rezaei	Convolutional Neural Networks	Basic ML Knowledge	YCB-Video Object Detection II

1. Objective and executive summary

The goal was to improve the Week 09 full-frame five-class Faster R-CNN detection pipeline by systematically changing confidence thresholds and class-aware non-maximum suppression (NMS), analyzing multiple-object frames, and interpreting precision–recall behavior. All reported Week 10 evaluation numbers below come from the uploaded, executed notebook.

Validation measure	Frozen baseline NMS 0.50 / score 0.50	Selected setting NMS 0.30 / score 0.30	Change
mAP@0.50	0.7193	0.7240	+0.0047
Macro-F1	0.6899	0.7074	+0.0175
Micro-F1	0.7017	0.7153	+0.0136
Micro precision	0.6608	0.6442	−0.0166
Micro recall	0.7480	0.8040	+0.0560

Central finding: lowering both thresholds improved validation recall and macro-F1 without retraining the detector, but reduced operating-point precision. The weakest class remains bleach_cleanser (recall 0.26; selected AP50 0.266). The result is validation-only, not an independently verified test-set improvement.

2. Relevant Main Quest concepts

CNN / detection concept	Where it was applied
Backbone + feature pyramid	COCO-pretrained MobileNetV3-Large/FPN extracts spatial features from complete RGB images.
Two-stage detection	Faster R-CNN creates region proposals, then predicts selected object class and bounding-box offsets.
Confidence vs localization	Prediction score controls which boxes remain; an IoU ≥ 0.50 class-aware match defines a true positive.
Non-maximum suppression	Class-aware ROI-head NMS removes overlapping, lower-scoring predictions before final scoring.
Precision–Recall / AP	Compute operating-point P/R/F1 and per-class AP50 from ranked predictions; interpret the trade-off.

3. Dataset, annotations and evaluation protocol

The same YCB-V BOP'19 frame-selection procedure as Week 09 was rebuilt from Hugging Face BOP archives (`ycbv_base.zip`, `ycbv_test_bop19.zip`). Five selected object IDs were mapped to contiguous detector labels: tomato_soup_can (4), sugar_box (3), master_chef_can (1), tuna_fish_can (6) and bleach_cleanser (12). The deterministic scene split uses up to 50 images per scene (seed 42).

Partition	Scene IDs	Frames	Selected GT boxes	Frames with ≥2 selected GT
Train	48, 49, 50, 53, 54, 56, 57, 58	400	584	200
Validation	52, 55	100	250	50
Previously seen test	51, 59	100	300	100

Labels were parsed from scene_gt.json / scene_gt_info.json. The primary box was bbox_obj (bbox_visib fallback), converted to clipped xyxy coordinates; instances required at least 0.35 visible fraction and a minimum width/height of 24 pixels. Across the validation split, each of the five selected classes has exactly 50 target objects.

3.1 Partial-label audit

Full RGB frames contain unselected YCB objects. The data summary recorded 1,150 other-YCB instances in training and 200 in validation. Additionally, 16 selected training objects failed filtering and 16 training frames contained no retained selected target. Thus, some apparent false positives overlap ignored regions; those examples cannot automatically be called model mistakes.

4. Implementation and training recovery

Component	Actual Week 10 implementation
Detector	Faster R-CNN / MobileNetV3-Large FPN; five foreground classes plus background (6 outputs).
Initialization	Fresh Week-09-compatible COCO-pretrained baseline; original Week 09 .pth was not available.
Model input	Full-frame RGB; internal detector resize min 480 / max 640 pixels.
Training	AdamW, LR 1e−4, weight decay 1e−4, StepLR (step 5, gamma 0.3); resumed FP32 with batch size 1.
Checkpoint rule	Highest validation mAP50; only this frozen checkpoint is used for the inference sweep.
Evaluation	Class-aware greedy one-to-one matching; IoU ≥ 0.50; AP ranking floor 0.05.

4.1 Recorded recovery history and incomplete training

Evidence in notebook	Observed result
Existing progress at resume	3 epochs completed; optimizer layout incompatible, so model weights restored and AdamW moments reset.
Epoch 4 completed	Train loss 0.6440; 2 invalid batches skipped; validation mAP50 0.7193 (best).
Epoch 5 completed	Train loss 0.6112; 3 invalid batches skipped; validation mAP50 0.7055.
Epoch 6 partial log	At least 2 invalid batches shown; no completed Epoch 6 summary in supplied notebook.
Model loaded for evaluation	Saved best checkpoint from Epoch 4; SHA-256 recorded in the notebook.

> Important: eight training epochs were configured, but the uploaded file does not demonstrate completion through Epoch 8. NaN classifier / box-regression losses still occurred on specific training samples in FP32; guarded batches were skipped rather than used for weight updates. This is a reliability limitation, not evidence of cleanly completed training.

5. Controlled validation experiments and results

5.1 Inference-only sweep

With the checkpoint fixed, Torchvision ROI-head NMS was rerun separately for IoU cutoffs 0.30, 0.50 and 0.70. For each NMS output, ten score thresholds (0.05–0.90) were evaluated on the same 100 validation frames: 30 combinations in total. Selection maximized validation macro-F1, with mAP50 and micro-F1 as tie-breakers. This avoids choosing solely for the most frequent/easiest class.

NMS	Confidence	Macro-F1	Micro-F1	Precision	Recall	mAP50
0.30	0.30	0.7074	0.7153	0.6442	0.8040	0.7240
0.30	0.40	0.7070	0.7161	0.6678	0.7720	0.7240
0.30	0.50	0.7023	0.7137	0.6825	0.7480	0.7240
0.50	0.40	0.6931	0.7004	0.6382	0.7760	0.7193
0.50	0.50	0.6899	0.7017	0.6608	0.7480	0.7193
0.70	0.60	0.6718	0.6903	0.6809	0.7000	0.6974

The 0.30 / 0.40 combination has slightly higher micro-F1 (0.7161), but the stated selection objective is macro-F1, so 0.30 / 0.30 is the selected setting. The NMS setting can affect AP50; score threshold changes the operating point, while AP50 is calculated from all retained predictions above 0.05.

5.2 Overall detection counts and localization

Validation metric	Baseline	Selected	Interpretation
True positives	187	201	14 additional matched detections
False positives	96	111	15 additional false positives
False negatives	63	49	14 fewer missed objects
Micro precision	0.6608	0.6442	Precision cost of higher recall
Micro recall	0.7480	0.8040	Higher coverage of selected targets
Mean IoU of baseline TPs	0.8403	—	Mean only over matched baseline TPs; not all GT boxes

5.3 Class-wise performance (all supports = 50)

Object	Base recall	Selected recall	Base F1	Selected F1	Base AP50	Selected AP50
tomato_soup_can	0.88	0.90	0.611	0.577	0.862	0.862
sugar_box	0.88	0.90	0.830	0.865	0.868	0.885
master_chef_can	0.72	0.96	0.632	0.716	0.607	0.607
tuna_fish_can	1.00	1.00	1.000	0.990	1.000	1.000
bleach_cleanser	0.26	0.26	0.377	0.388	0.259	0.266

The largest class-wise recall gain was for master_chef_can (0.72 → 0.96). Lower confidence also created additional detections that reduced tomato_soup_can F1 (0.611 → 0.577). Bleach_cleanser remains the main detection bottleneck despite a small AP50 increase (0.259 → 0.266).

6. Saved precision–recall and AP50 outputs

The following figures are extracted from the uploaded executed Notebook, not recreated from invented measurements. Precision–recall plots use the interpolated precision envelope from ranked predictions at IoU 0.50 and a confidence floor of 0.05. Vertical steps are expected for discrete detections and small class support.

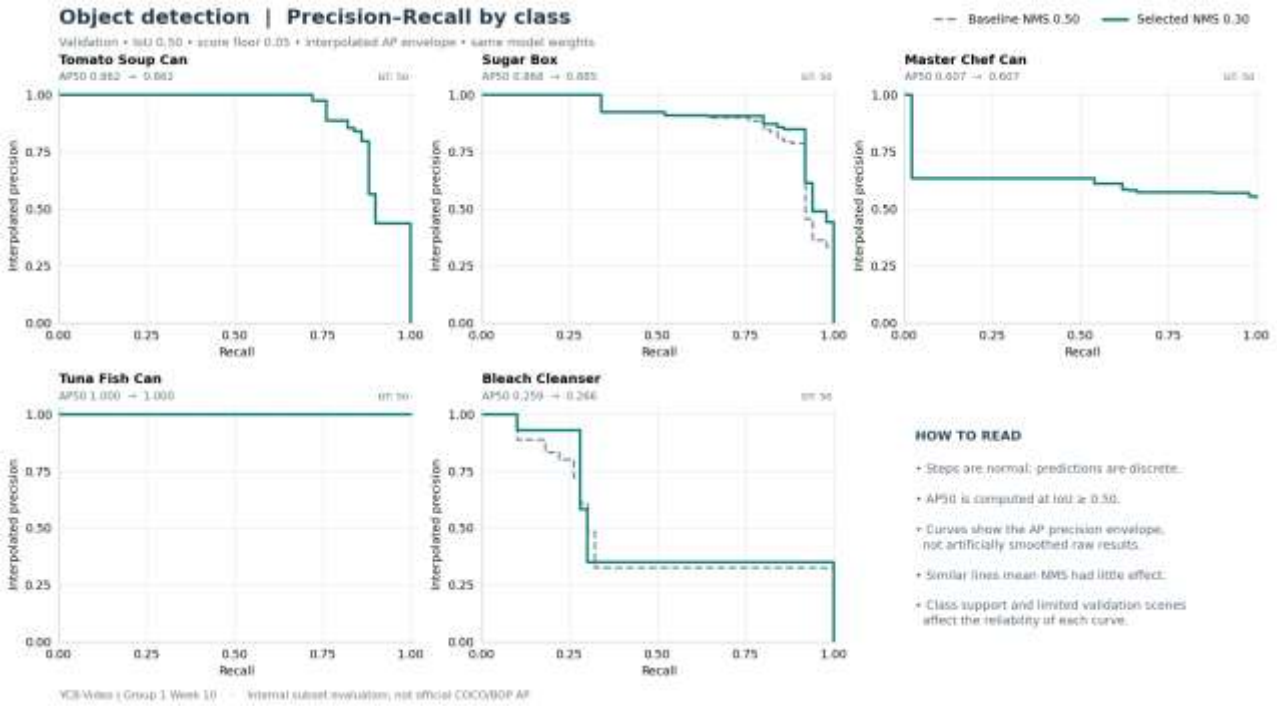


Figure 1. Actual class-wise validation precision–recall envelopes: baseline NMS 0.50 versus selected NMS 0.30. Each class has 50 labeled GT instances.

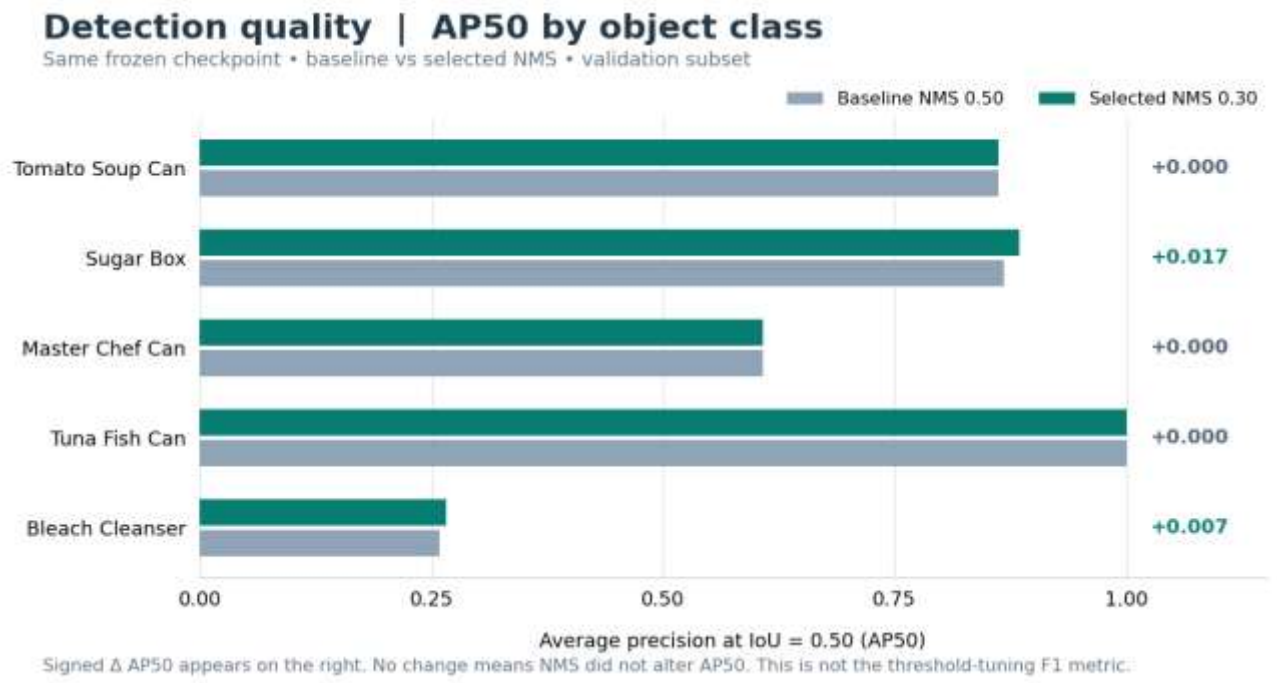


Figure 2. AP50 by class from the executed validation comparison; the largest improvements are sugar_box (+0.017) and bleach_cleanser (+0.007).

7. Confidence, NMS and operating-point trade-offs

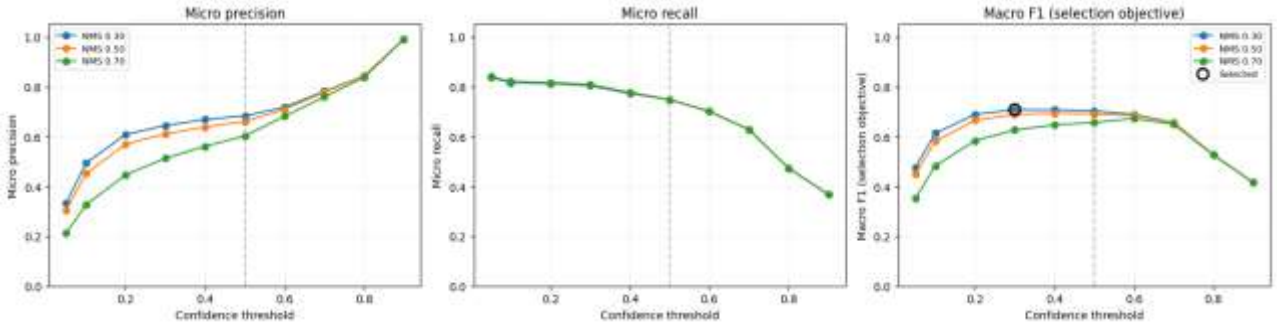


Figure 3. Validation micro precision, micro recall and macro-F1 across ten confidence thresholds, for three NMS values. The selected macro-F1 point is marked.

As confidence increases, predictions are removed: precision generally rises while recall falls. The selected score of 0.30 achieves greater recall but admits more false positives than the baseline 0.50 cutoff. The setting was chosen through validation macro-F1 rather than maximum precision.

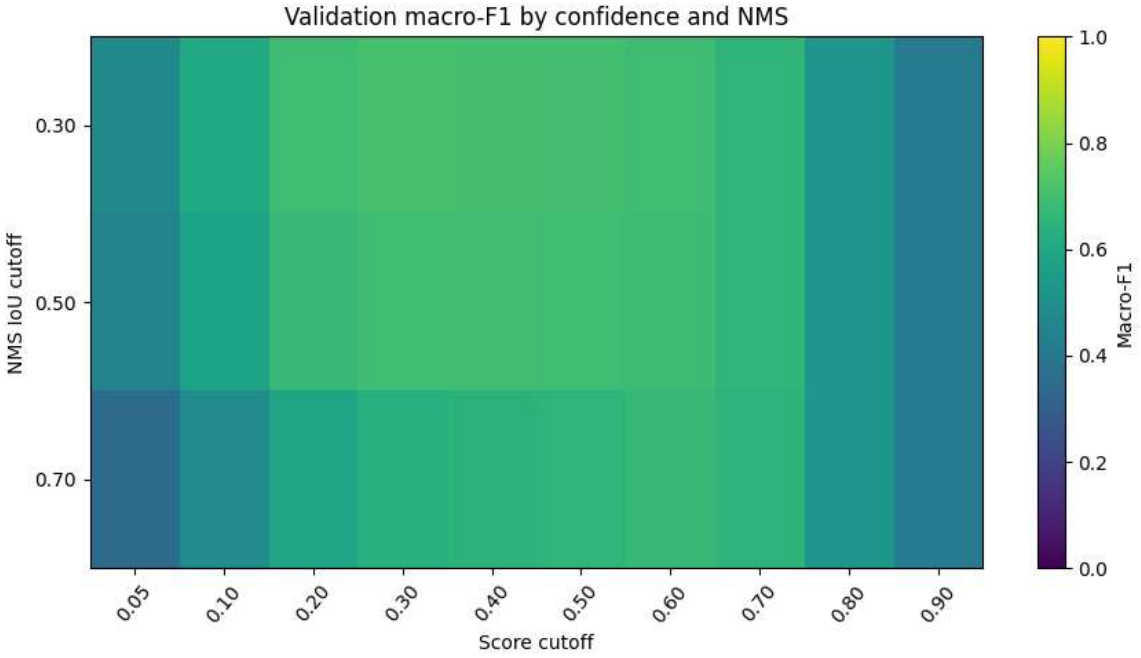


Figure 4. Thirty-setting validation macro-F1 heatmap. The search is limited to three class-aware NMS cutoffs and ten confidence thresholds; axes use different concepts (overlap IoU versus confidence).

NMS changes are modest at the aggregate AP50 level: NMS 0.50 gives mAP50 0.7193 and NMS 0.30 gives 0.7240. The selected configuration is a useful operating-point adjustment, not a substitute for fixing underlying missed objects or mislabeled regions.

8. Multiple-object evaluation and false-positive analysis

8.1 Multi-object and one-target frames

The 100 validation frames divide evenly into 50 frames with at least two retained selected targets and 50 with zero or one (in this actual split, each of the latter contains exactly one). Subgroup statistics cannot be compared as if they had identical target composition.

Validation group	Setting	Frames	Micro-P	Micro-R	Micro-F1	TP / FP / FN
≥ 2 selected GT	Baseline	50	0.668	0.685	0.677	137 / 68 / 63
≥ 2 selected GT	Selected	50	0.668	0.755	0.709	151 / 75 / 49
≤ 1 selected GT	Baseline	50	0.641	1.000	0.781	50 / 28 / 0
≤ 1 selected GT	Selected	50	0.581	1.000	0.735	50 / 36 / 0

The selected setting improves micro-F1 on multiple-target frames (0.677 → 0.709; +14 true positives, −14 false negatives), but worsens it on one-target frames (0.781 → 0.735) by generating eight extra false positives with no additional correct detections.

8.2 False-positive audit – heuristic categories

Category of unmatched prediction	Baseline FP	Selected FP
Possible wrong predicted class	48	50
Other / background unmatched FP	32	34
Possible overlap with ignored/partial-label region	13	24
Possible localization error	3	3
Total	96	111

These are geometry-based hypotheses, not manually verified error diagnoses. The 24 selected-setting predictions overlapping ignored regions remain counted as FPs in the headline metrics; excluding them without review would change the evaluation definition. No duplicate-same-class category appeared in the recorded aggregation.

8.3 Specific frame-level improvements and regressions

Scene / frame	Change in (FP + FN)	Reading
55 / 1567, 1625, 1543	−2 each	Two fewer combined mistakes in each selected frame.
55 / 793	+2	Two additional combined mistakes after selection.
52 / 575, 604, 625, 643, 678	+1 each	One extra FP with no gain in GT recall.

9. Representative real-image predictions

All images in Figures 5–6 are screenshots of the actual validation visualizations saved inside the executed Notebook. Each row compares ground truth (left; gray dashed = ignored region), baseline (middle) and tuned output (right). The selected images intentionally include both improvements and regressions.



Figure 5. Validation scene 55 examples: frames 1567 and 1625 have fewer errors, while frame 793 regresses.

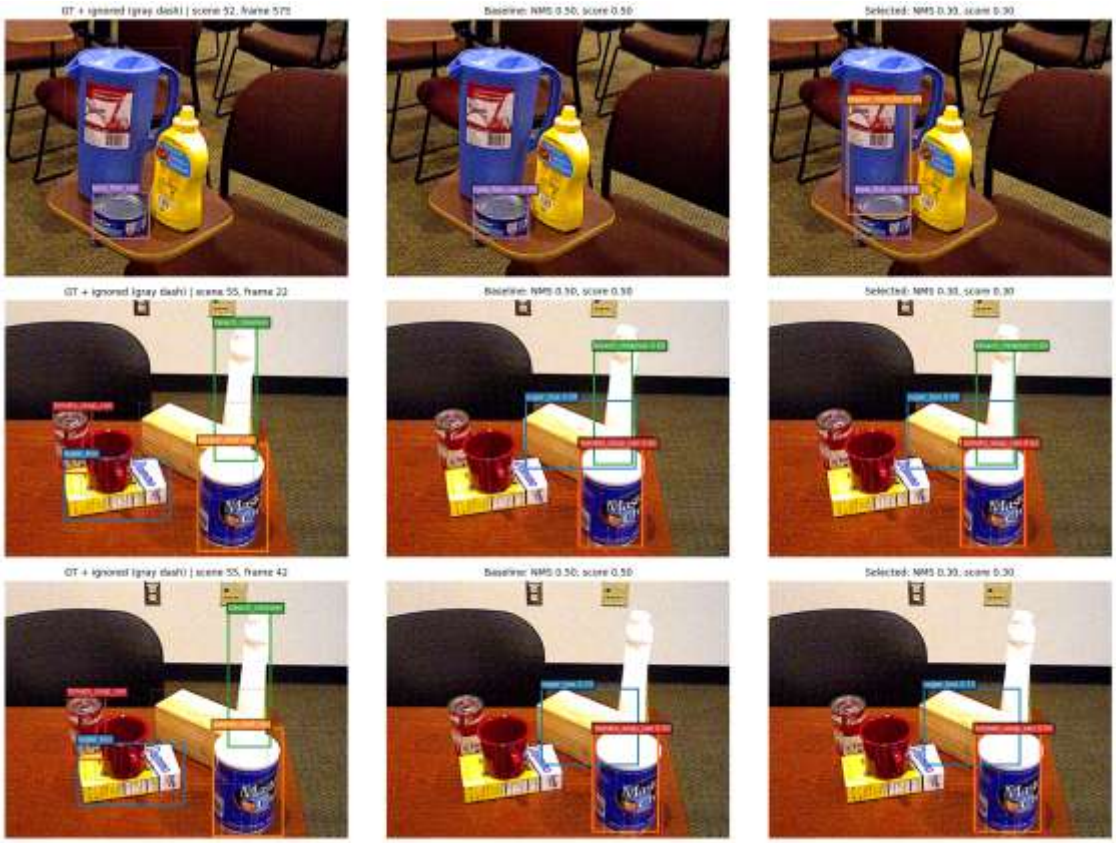


Figure 6. Scene 52 frame 575 and scene 55 frames 22 and 42: further correct and ambiguous detections.

10. Failure analysis and limitations

- Weak bleach_cleanser generalization: on 50 validation GT boxes, recall stayed at 0.26 for both settings; low AP50 (0.266) indicates the class remains difficult even when ranking predictions by score.
- Class confusion and clutter: the FP audit classified 50 of 111 selected-setting FPs as possible wrong-class predictions; close object appearance, occlusion and clutter plausibly contribute, but were not isolated in controlled causal tests.
- Precision–recall compromise: the new operating point gains 14 TPs and loses 14 FNs, but introduces 15 FPs; this is not an unqualified improvement for every task.
- Incomplete annotations: 200 non-selected YCB instances appear in the validation images, and 24 selected-setting FPs overlap ignored regions according to the heuristic audit.
- Limited independence: tuning used only scenes 52 and 55. Prior-week scenes 51/59 were deliberately not retested here, and the notebook does not include an untouched external test split.
- Training instability: the recovered training path included NaN loss components and optimizer reset. Best model at epoch 4 was evaluated, but a complete eight-epoch stable run is not evidenced. The historical Week 09 validation mAP50 of 0.6970 came from different weights and is not a controlled Week 10 improvement baseline.

11. YOLO-style concepts – comparison, not a YOLO experiment

Faster R-CNN is a two-stage system: a proposal mechanism identifies candidate regions, and ROI heads classify/refine them. A YOLO-style detector uses one-stage prediction heads over feature maps. Both may use confidence filtering and NMS depending on the implementation. This week did not train or benchmark a YOLO model, so no speed or accuracy superiority is claimed.

12. Conclusion and next-step recommendation

Week 10 demonstrated a controlled inference-only improvement on a frozen, newly trained Week-09-compatible detector: validation mAP50 increased from 0.7193 to 0.7240 and macro-F1 increased from 0.6899 to 0.7074 at NMS 0.30 and score 0.30. The model detects multiple selected objects more completely, especially master_chef_can, but its lower precision and persistent bleach_cleanser misses make the outcome mixed rather than uniformly successful.

For Week 11, carry forward this five-class data pipeline and frozen validation protocol, inspect segmentation mask coverage/quality before choosing a mask-based model, measure mask IoU per class, and document boundary failures. Before making stronger detector claims, separately audit ignored-region overlaps, repeated NaN samples, and complete an independent test on genuinely unseen scenes.